



Quantum Computing

State of Affairs Since the August Edition

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About This Report

This is an experimental, AI-driven research project from SRL. SRL is independent, with no outside funding and no sponsors. It uses several AI models from competing labs, running on different platforms. One model gathers the current claims on a topic, a different one checks each claim against its source, and disagreements are kept, not smoothed over. The aim is an even-handed account that no outside party has shaped. AI makes mistakes, and so does the system built around it. We find and fix some in every cycle, so the methods keep changing and the content should improve over time. Updates come at least quarterly, sooner if something major happens.

This report covers quantum computing from the start of September to 4 October, the stretch since the last full edition. It is written to stand alone. A reader who skipped the earlier editions will find the background they need at the end.

The report opens with the handful of developments that change the picture. It then turns to government money, followed by the companies, their partnerships and their private funding. The public markets come next, then the machines themselves and the error correction that every machine depends on. Theory follows, with its sharpening account of what noisy hardware can and cannot do. The claims of useful results close the new material.

The last part steps back. It describes where the field stood before September, so the month's news can be read against it.

1. What Matters Most

A month of news holds a few developments that change the picture and many that only refresh it, and this section names the ones that change it.

The month had a clear shape. Governments signed what they had promised and set new targets. One European company stepped back from a defence programme. Theorists drew a sharper line around what noisy machines can do, just as a team reported break-even on a real chemistry problem.

Start with Germany. Its research ministry is funding a programme called NFQC-1k with about 122 million euros over up to five years [1]. A German ion-trap start-up leads the consortium, and its members include



two universities, Germany's national metrology institute, NXP, Forschungszentrum Jülich and AQT [1]. The target is a demonstrator with at least 1,000 physical qubits and 50 logical qubits, plus a pilot production line for ion-trap processors [1]. That is a national bet on one hardware approach, with a logical-qubit number attached.

France moved the other way for one company. Pasqal said on 28 September that it is leaving France's defence quantum programme after taking part in its first phase [2]. The company says it met all of that phase's technical objectives [2]. Two French government agencies have asked it to propose a civil research programme toward fault-tolerant neutral-atom computing instead [2]. Whether that civil track is funded is the open question.

The American money hardened. In May, reports had put GlobalFoundries down for about \$375 million in CHIPS Act funding and several quantum firms for roughly \$100 million each [3]. GlobalFoundries finalized its \$375 million award on 21 September, for quantum research, cryogenic electronics and advanced packaging [4]. PsiQuantum, Rigetti, Quantinuum and D-Wave each signed awards worth up to \$100 million [5] [6] [7] [8]. The Department of Energy then opened a competition worth up to \$215 million for fault-tolerant machines with at least 100 logical qubits by 2028 [9].

From money to theory. A new proof shows that a classical computer can sample, in polynomial time, the output of any geometrically local circuit under steady depolarizing noise once the circuit is deep enough [10]. Depolarizing noise is the simplest error model, in which a qubit is randomly scrambled at a fixed rate. The depth threshold depends only on that rate, not on how many qubits there are [10]. Beyond that depth, a classical computer can keep up with noisy hardware that lacks error correction.

In the same week, a team including Mitsubishi Chemical and IBM Research in Tokyo reported break-even on a 120-qubit simulation of electron dynamics in a ring molecule [11]. The processor's wall-clock time matched estimated classical runs on tens to hundreds of compute nodes [11]. The authors say plainly that this is not quantum advantage [11]. A result framed that carefully is itself a sign of how the field has learned to argue.

The listings kept moving too. EigenQ's merger with a listed shell company went to a public registration filing [12]. IonQ raised its full-year revenue guidance to \$450 million to \$460 million after buying SkyWater [13]. Pasqal, newly listed, reported its first half-year results [14].

2. Governments and Public Money

Public money now sets the pace of this field, and the month added new programmes in several countries and one military alliance.

The headline programmes are covered above. Germany's NFQC-1k and the American CHIPS awards change who pays for the next generation of machines [1] [4]. Beyond them, several smaller commitments filled in the map.

The Department of Energy's competition is called Quantum Genesis Q, and it is aimed at private companies [9]. The bar it sets is a scientifically relevant fault-tolerant machine with at least 100 logical qubits by 2028 [9]. That is about twice the 48 error-corrected logical qubits Quantinuum reported in March [15].

National laboratories took on partners as well. PsiQuantum is working with Brookhaven National Laboratory on fault-tolerant algorithms, using its Construct software platform [16]. In Albuquerque, a 35,000-square-foot facility called the Roadrunner Quantum Lab opened in late September [17]. It links Sandia and Los Alamos with private companies, including Diraq, IonQ and QuEra [17].

Other governments joined in. Australia awarded \$12.3 million in grants to six quantum companies on 1 October, with Silicon Quantum Computing receiving the largest at \$3.6 million [18]. NATO published the public summary of its quantum technology roadmap the same day, with timelines and standards work across computing, sensing and communications [19]. India's National Quantum Mission co-led a Series A1 round of about \$21 million for QNu Labs [20].

France was the exception. Pasqal left its defence programme, as described above [2]. The civil programme it has been asked to propose would be its next government track, if approved.

3. Companies, Partnerships and Private Money

Company announcements are promises as much as results, so this section separates what was built or signed from what was only planned.

Signed partnerships came first. Diraq will open a research laboratory in Albuquerque under an agreement with the Roadrunner Quantum Lab [21]. The company says the lab will open by the end of the year with a dedicated measurement engineering team [21]. It follows Diraq's August roadmap, which targets more than 2 million physical qubits on one chip by 2031 [22]. Xanadu and Bluefors, a maker of ultra-cold refrigerators, agreed a multi-million-dollar collaboration on a cryogenic prototype for utility-scale machines [23].

Roadmaps were the second kind of news, and they are company statements, not results. Xanadu targets more than 1,000 logical qubits by 2031 [24]. Its milestones run from a qubit factory in 2026 and 2027 to a quantum data centre by 2030 [24]. IonQ's new roadmap has a 256-qubit system reaching customers in 2027 [25]. A 10,000-qubit system is to demonstrate fault tolerance in 2027 and go on sale in 2028 [25].

D-Wave, best known for its annealing machines, took a step into gate-model computing. On 1 October it opened a beta programme for a gate-model simulator built around its dual-rail superconducting design [26]. BBVA, Florida Atlantic University and the Jülich Supercomputing Centre are among the participants [26].

Private rounds stayed small next to public money. Mesa Quantum raised \$11.8 million for chip-scale quantum sensors [27]. IBM Ventures invested in BQP, a physics acceleration firm, bringing its total funding to \$8 million [28]. A start-up closed a \$2.4 million pre-seed round for quantum control software [29].

4. Public Listings and Markets

Public markets have become the field's second funding channel, and this section follows the listings and the numbers listed companies report.

The pending deals advanced. EigenQ and its listed shell company, Silicon Valley Acquisition Corp., filed their registration statement publicly on 28 September [12]. It has not yet been declared effective [12]. The

combined company would be called EigenQ Holdings and trade on Nasdaq under EIGQ, with warrants under EIGQW [12].

Two traditional offerings moved forward. SEEQC amended its Nasdaq registration on 22 September, with no price range set [30]. It had first filed in June as a superconducting company putting control electronics on the chip [31]. In China, the photonic company TuringQ began the listing process in August, seeking the country's first quantum listing [32]. A second Chinese firm, Qboson, registered for the same process in July [30].

The newest listing reported for the first time. Pasqal's first-half revenue was 4.9 million euros, up 14 percent, against an operating loss of 59.2 million euros [14]. Its cash was about 312.9 million euros after the merger closed [14]. A registration statement it filed on 9 September took effect on 18 September, followed by a prospectus and an amendment [33].

The older names showed the range of fortunes. IonQ's higher guidance counts five months of SkyWater, while its standalone guidance stays at \$280 million to \$290 million [13]. The New York Stock Exchange removed IonQ's old warrants from listing on 30 September [34]. At the other end, two pure-play stocks traded under five dollars in mid-September [35].

5. The Machines

Hardware news splits by approach, because superconducting circuits, trapped ions, neutral atoms, photons and silicon spins are different physics, not variations on one design.

IBM had the busiest month among the superconducting builders. Its Nighthawk r2 processor, released on 31 August, adds fast reset to each of its 120 qubits [36]. That lets it run more than 100,000 circuits a second, 25 times the throughput of its Heron chips [36]. It also estimates results accurately on circuits of more than 7,500 two-qubit gates, up from 5,000 [36].

The same chip carries an advantage claim. On 61 qubits, it generated a million random-circuit samples in 19 seconds [37]. Its fidelity estimate was 2.3×10^{-3} , and the authors put the classical cost at 1.2×10^{27} operations, over 100 years on the Frontier supercomputer [37]. Separately, IBM joined two large cryogenic modules that reached 4 kelvin in under five days and then cooled below 15 millikelvin [38].

Other superconducting work stayed in the laboratory. MIT proposed an "arm qubit" design that separates storage from interaction, with faster operations in simulation [39]. MIT and collaborators also built a qubit entirely from stacked layered materials, with an average energy-relaxation time of 55 microseconds [40].

The trapped-ion side added a machine and an idea. eleQtron and Forschungszentrum Jülich launched a ytterbium-ion computer controlled by microwaves, for hybrid use [41]. Maryland and NIST researchers proposed ion-trap gates that rely less on real-time classical control [42]. An earlier Oxford result showed fourth-order squeezing in a single trapped ion [43].

Silicon and germanium spins posted strong numbers. A Chinese team ran geometric gates on a germanium hole-spin qubit with fidelities above 99 percent across a wide range of drive strengths, peaking above 99.9 percent [44]. RIKEN initialised a pair of silicon spin qubits without measurement feedback, reaching a median

success probability of 99.4 percent [45]. Quantum Motion reported 99.92 percent readout fidelity from a compact sensor on a 300-millimetre wafer [46].

Photonics added a design, not a device. Bristol researchers published a blueprint for low-depth photonic computing with quantum-dot light sources, with logical clock cycles in microseconds [47].

6. Error Correction

Error correction spreads one protected logical qubit across many unreliable physical ones, and its cost decides what a useful machine will cost to build.

The month's most original hardware result came from Alice & Bob. Cat qubits store information in a pattern of light in a microwave cavity, built so that one kind of error almost never happens. The company stabilized them with a plain DC voltage bias instead of a microwave drive [48]. It reached a two-photon exchange rate of 3 MHz and could switch between one, two and four photon exchanges on the same chip [48].

Decoding improved sharply on paper. A decoder is the classical software that reads error signals and decides what to fix. A postselection method called argument reweighting, published in Physical Review Letters, cuts the logical error rate of IBM's gross code, its leading high-rate code, by nearly two orders of magnitude [49]. It throws away only 1.44×10^{-5} of runs to do so [49]. A soft beam search decoder claims up to 580 times error suppression at a physical error rate of 10^{-3} , and doubled logical lifetimes in trapped-ion experiments [50].

The newer high-rate codes found new homes. A three-dimensional layout of the gross code on neutral atoms packs logical qubits about four times denser and reads out errors about twice as fast [51]. A surgery method for neutral atoms runs 33 logical measurements in parallel about twice as fast as earlier schedules [52]. Layer codes reached a circuit-level threshold of $5.06(7) \times 10^{-3}$ in simulation, at about five times the qubit cost of a surface code [53].

Surface codes gained a new platform. USC and Quantum Elements showed below-threshold scaling on IBM's Heron processors [54]. A compiler pass for lost atoms cut logical error rates by up to 5.3 times in 35 of 48 simulated settings [55]. A runtime for preparing magic states, the ingredient that makes protected computation universal, cut preparation time by up to 39.3 percent [56].

Unresolved Conflicts

One disagreement sits here. IonQ's own earlier preprint had its single-CPU decoder handling 408 logical qubits and over 1 million T gates at under 0.3 percent overhead. A September news report gives the same 408 qubits but over 31.5 million operations at 0.02 percent overhead [57]. The two may describe different runs, or the news report may have the figures wrong. A dated update from IonQ naming the workload would settle it.

7. Theory: What Noise Allows

Theory says what quantum computers can and cannot do faster than ordinary ones, and this month it focused on noise.

The noise results fit together like a map. The new classical sampler covers noisy local circuits of any shape past a fixed depth [10]. A second paper shows that noisy shallow circuits on a line give no advantage over a simple class of classical circuits [58]. A third proves that noisy circuits on a flat grid, below a noise threshold, do beat that same class unconditionally [59]. Between them, the three papers tie the question of advantage to geometry, depth and noise rate.

Two papers question the error models themselves. Gil Kalai revisited his 2006 conjecture that entangled qubits carry a joint, fully scrambling noise component at a rate comparable to two-qubit gate errors [60]. He says it is still open but can now be tested on current devices [60]. A separate preprint argues that one layer of noise at any small rate breaks Shor's algorithm, and proposes a classical polynomial-time factoring method [61]. That is an extraordinary claim in an unreviewed preprint.

On the constructive side, the results were steadier. One paper proves a quantum advantage for approximate optimization using decoded quantum interferometry, on a folded version of a known problem [62]. Another proves an advantage for QAOA, a standard quantum optimization algorithm, on regular graphs without short cycles [63]. A third pins down the exact constant for quantum ordered search [64].

Fault tolerance gained theory too. A unified spacetime framework recovers several known syndrome extraction methods as special cases [65]. A result on good quantum locally testable codes depends on objects the authors have not yet built [66]. One preprint cut the cost of a secp256k1 point-addition circuit used in Shor's attack on elliptic curves by 86.1 percent in space-time cost [67]. A new error-mitigation method cuts sampling cost by up to a thousand times against a standard approach [68].

8. Claims of Useful Results

This section pairs each claim of useful quantum work with its conditions, because the conditions decide what the claim is worth.

The chemistry break-even result is the month's most careful claim. It embeds a 12-qubit benzene model and a 120-qubit ring molecule onto loops of IBM processors [11]. Compression with tensor networks keeps the circuits short [11]. The classical comparison assumes near-ideal parallel scaling, and the paper frames the outcome as break-even, not advantage [11].

Hybrid systems became concrete. Japan's Fugaku supercomputer coordinated a Quantinuum machine and an IBM machine in one ground-state energy workflow [69]. A Cornell group used Grover's algorithm on one superconducting qubit to detect signals of unknown frequency [70]. It beat the non-Grover baseline above 10 MHz of bandwidth [70]. For wider bands and stricter accuracy, it cut the signal needed by more than ten times [70].

Benchmarks and smaller claims filled in. A new cross-platform benchmark scored Quantinuum's Helios at 1504, Google's Willow at 216 and an IBM machine at 204 [71]. Its authors project five orders of magnitude of growth before problems like factoring RSA-2048 come into reach [71]. In finance, sixteen features from an eight-qubit circuit lifted a credit model's F1 score from 0.462 to 0.517 [72].

Unresolved Conflicts

One disagreement sits here. IBM's University of Chicago collaboration says no classical computer can produce its circuit's sample distribution in reasonable time [73]. A Singapore team reports computing exact amplitudes for all 2,051 of its samples in 37.3 minutes on 256 GPUs [74]. The rebuttal is a single, uncorroborated report. A peer-reviewed reply from either side, or an independent rerun, would settle it.

9. Where Things Stood Before This Update

Readers who skipped the earlier editions need the picture this month's news changed, and this section gives it.

Before September, the field's central scientific question had been answered. Google's Willow processor ran the first quantum memory below the surface-code threshold [75]. Going from a distance-5 code to a distance-7 code cut the logical error rate by a factor of 2.14 [75]. Adding qubits finally made the protected qubit better, not worse.

The price of that protection was falling. IBM's gross code encodes 12 logical qubits in 288 physical ones, about a tenth of what a surface code would need [76]. Quantinuum's 98-qubit Helios ran up to 48 error-corrected logical qubits [15]. Caltech's Endres group held the size record, with an array of 6,100 atoms [77].

Money had moved to governments and public markets. Venture funding for quantum start-ups was on track to finish 2026 below the 2025 record [78]. In May, the Department of Commerce announced letters of intent for about \$2.013 billion in CHIPS Act incentives [79]. IBM was expected to receive about \$1 billion, and reports said the government would take equity stakes [3].

Listings were the other channel. Quantinuum raised \$1.68 billion in June at a valuation of \$17.63 billion, the largest pure-play quantum listing to date [80]. Pasqal began trading on Nasdaq on 28 August with about \$360 million at closing [81]. The earlier wave of listings had delivered about twelve cents on each projected revenue dollar [82].

Claims of advantage were met by rebuttals. In July, IBM coordinated three advantage papers, which carried different weight [83]. A classical method running on a laptop had already matched D-Wave's 2025 supremacy result on spin glasses [84]. That pattern of claim and answer is still running, as the Singapore reply above shows.

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